Marketing Analytics

Customer Data and Measurement

By Nathan Yang

Session 02

A Customer-Activity Decision

- An mHealth manager wants to understand reduced recorded use.
- The comparison follows the same eligible customers over time.
- A missing record can reflect nonuse, incomplete observation, or data coverage.

A Customer Table

- Begin with the eligible-customer roster.
- Aggregate events to the customer-period level.
- Left join the summaries to the roster.
- Fill zero only when absence has the defined meaning of no recorded event.

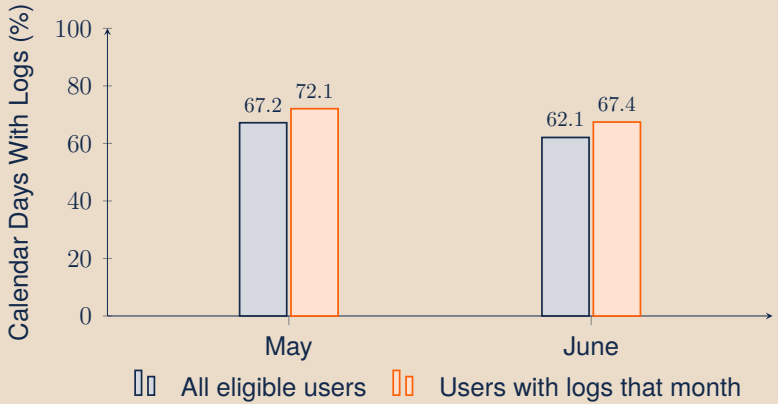
Two Activity Measures

Let D_i be observed logging days and T the calendar days in a common window.

$$L_i = \frac{D_i}{T}, \qquad \bar{L} = \frac{1}{N}\sum_{i=1}^{N} L_i$$

Each eligible customer receives equal weight, including customers with no recorded days.

Conditional and Fixed Denominators



mHealth activity depends on who remains in the denominator.

Repeated Totals

- An item row can contain a copied daily total.
- Five copies of 1,132.79 sum to 5,663.95.
- The item-calorie sum on that example day is 1,132.7852.

Structural Errors and Review Flags

- A duplicate daily key violates a one-row-per-day contract.
- Equal item descriptions can represent repeated servings.
- An extreme value warrants review under a stated rule.

A Fixed Customer Denominator

Synthetic counts follow all four eligible customers.

```
days = {'A': 20, 'B': 10, 'C': 0, 'D': 0}
fixed = sum(days.values()) / (4 * 30)
active = [x for x in days.values() if x > 0]
conditional = sum(active) / (len(active) * 30)
assert round(fixed, 4) == 0.25
assert round(conditional, 4) == 0.50
```

The fixed-roster rate is 25%. The active-user rate is 50%.

Cohorts and Customer Age

- Group customers by a defined entry period.
- Compare the same time since entry.
- Mark later outcomes as unobserved when the horizon has not elapsed.

Retention Has a Triangular Support

	At Entry	Month 1	Month 2	Month 3
January	100%	60%	45%	40%
February	100%	70%	50%	Not yet observed
March	100%	80%	Not yet observed	Not yet observed

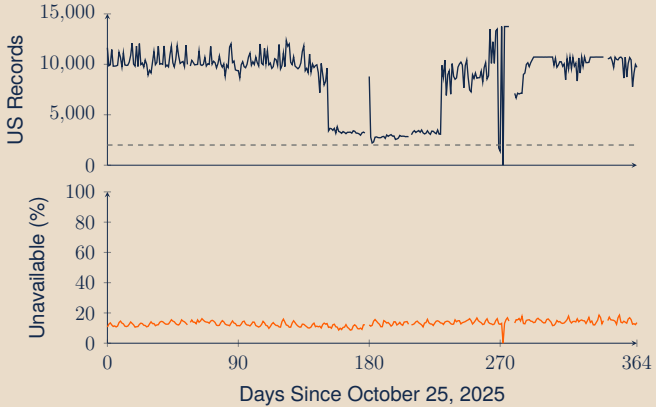
Each cohort begins with 100 customers.
Only observed customer ages enter the retention denominator.

Synthetic cohort structure. Blank cells have no elapsed follow-up.

McBroken as a Measurement Case

- The records describe ice-cream ordering availability.
- A missing snapshot has an unknown status.
- Restaurant coverage and status recognition can change over time.

Coverage Changes the Percentage



McBroken comparisons preserve the observed-store denominator.

Identity Across Files

- A restaurant address is a candidate key that requires checks.
- Standardizing spelling can repair formatting differences.
- Overaggressive matching can merge separate locations.

Coding Exercise: A Data Audit

- Construct a roster-first activity table.
- Report key uniqueness, unmatched rows, and zero-event customers.
- Compare the fixed and conditional activity measures.

AI Review of a Parser

- Supply the allowed statuses and a tiny labeled example.
- Require an explicit unknown category for new statuses.
- Compare parsed totals with an independently calculated source total.

A Data Dictionary for Decisions

- State the unit, key, timing, and units of each variable.
- Record source versions and transformation rules.
- Separate observed values, calculated quantities, and imposed assumptions.

Privacy and Collection Choices

- Collect the fields needed for the stated decision.
- Retain source permissions and deletion or retention conditions.
- Check whether a linkage increases the risk of identifying a person.

Measurement Discussion

- Could recorded activity rise because low-activity users disappear?
- Would deleting unmatched days change the target population?
- What evidence would establish a mechanical failure?